SAMADH HIGHER SECONDARY SCHOOL
KHAJANAGAR, TRICHY-20

QUARTERLY EXAMINATIONS (2026-27)

CLASS: XII **MARKS: 70**

SUBJECT: CHEMISTRY **TIME: 3Hrs**

General Instructions:

Read the following instructions carefully.

1. There are 33 questions in this question paper with internal choice.
2. SECTION A consists of 16 MCQs carrying 1 mask each.
3. SECTION B consists of 5 short answers question carrying 2 marks each.
4. SECTION C consists of 7 short answers question carrying 3 marks each.
5. SECTION D consists of 2 case based questions carrying 4 marks each.
6. SECTION E consists of 3 long answer questions carrying 5 marks each.
7. All questions are compulsory.
8. Use of log tables and calculators is not allowed.

SECTION – A 16x1=16

1. Identify the IUPAC name of the given compound $CH_3 - \underset{\displaystyle Cl}{\underset{|}{CH}} - C_6H_5$

A) 1 – Chloro – 1 – Ethyl benzene

B) 1 – Chloro – 1 – methyl benzene

C) 1 – Chloro – 1 – Phenyl ethane

D) 2 – Chloro – 2 – Phenyl ethane

2. Find 'X' in the given chemical equation

$$CH_3 - CH_2 - CH = CH_2 \xrightarrow[\text{Peroxide}]{\text{H Br}} \text{'X'}$$

A) $CH_3 - CH_2 - \underset{\displaystyle H}{\underset{|}{CH}} - \underset{\displaystyle Br}{\underset{|}{CH_2}}$

B) $CH_3 - CH_2 - \underset{\displaystyle Br}{\underset{|}{CH}} - \underset{\displaystyle H}{\underset{|}{CH_2}}$

C) $CH_3 - CH_2 - \underset{\displaystyle H}{\underset{|}{CH}} - \underset{\displaystyle H}{\underset{|}{CH_2}}$

D) $CH_3 - CH_2 - \underset{\displaystyle Br}{\underset{|}{CH}} - \underset{\displaystyle Br}{\underset{|}{CH_2}}$

3. Identify the product

$$\begin{array}{c} CHO \\ | \\ (CHOH)_4 \\ | \\ CH_2OH \end{array} \xrightarrow{\text{HCN}} \text{Product}$$

A) $CH_3\ CH_2\ CH_2\ CH_2\ CH_2\ CH_3$

B) $\begin{array}{c} CHO \\ | \\ (CHOH)_4 \\ | \\ CH_2CN \end{array}$

C) CH_2CN
$(CHOH)_4$
CH_2OH

D) $CH(CN)(OH)$
$(CHOH)_4$
CH_2OH

4. Deficiency disease caused by vit-k is

A) scurvey
B) Increased blood clotting time
C) Rickets
D) Convulsions

5. Phenols are more acidic than alcohols due to

A) Inductive effect
B) Hydrogen bonding
C) Resonance stabilization of phenoxide Ion
D) Higher molecular mass

6. Which of the following is a strongest acid.

A) Phenol
B) o-cresol
C) p-cresol
D) p-nitrophenol

7. The boiling point of an azeotropic mixture of water-ethanol is less than that of water and ethanol. The mixture shows

A) No deviation from Raoults law
B) $+^{ve}$ deviation from Raoults law
C) $-^{ve}$ deviation from Raoults law
D) None of the above

8. Which of the following aqueous solutions will have highest boiling point.

A) 1.0M KCl
B) 2M KCl
C) 1.0M K_2SO_4
D) 2M K_2SO_4

9. A solution of NaCl shows higher osmotic pressure than glucose solution of same concentration. Predict the reason.

A) NaCl has higher molar mass
B) NaCl dissociates into ions
C) NaCl associates as a dimer
D) Glucose dissociates into ions

10. A solution freezes at a lower temperature than pure solvent. This is due to

A) Increases in vapour pressure
B) Decreases in vapour pressure
C) Decreases in osmotic pressure
D) No change in pressure

11. An alcohol that does not undergo oxidation but gives alkene on heating is

A) Primary Alcohol
B) Secondary Alcohol
C) Tertiary Alcohol
D) Allylic Alcohol

12. Phenol reacts with NaOH but ethanol does not because

A) Phenol is less soluble
B) Phenol is less Acidic
C) Phenol is more Acidic
D) Ethanol is more reactive

Q. No 13 – 16, Choose the correct options.

A) Both A and R are true and R is the correct explanation of A.

B) Both A and R are true but R is not the correct explanation of A.

C) A is true but R is false.

D) A is false but R is true.

13. **Assertion (A) :** Lucas test distinguishes primary, secondary and tertiary alcohols.

    Reason (R) : Tertiary alcohols form carbocations more easily than other alcohols.

14. **Assertion (A) :** Glycogen is stored in animals instead of starch.

    Reason (R) : Glycogen is more highly branched than starch.

15. **Assertion (A) :** Vitamins are essential in small amounts in diet.

    Reason (R) : Vitamins cannot be synthesised in adequate amount by the body.

16. **Assertion (A) :** Denaturation of proteins leads to loss of biological activity.

    Reason (R) : Denaturation disrupts the primary structure of proteins.

SECTION – B **(5x2=10)**

17. Write the suitable chemical reactions for the following:

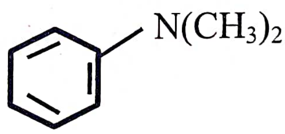
    (i) Finkelstein reaction (ii) Swarts reaction

18. Convert the following.

    (i) Aniline into Iodobenzene (ii) Chloromethane into Ethane

19. Find the IUPAC names of

    (i) $CH_3-\underset{\underset{\displaystyle H}{|}}{N}-CH_2CH_2CH_3$ (ii)



20. Write suitable chemical reactions of Gabriel phthalimide synthesis.

21. A DNA sample is found to have 30% Adenine. Calculate the percentage of Guanine in the sample.

(OR)

Give the deficiency disease of

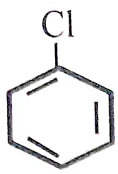
(i) Vit – D (ii) Vit – B_{12}

SECTION – C **(7x3=21)**

22. Write suitable reactions.

    (i) Kolbe's reaction (ii) Reimer Tiemann reaction

23. Complete the following reactions.

(i)  $+ C_2H_5Cl \xrightarrow{\text{Anhyd. } AlCl_3}$

(ii)  $\xrightarrow[\Delta]{\text{Conc. } H_2SO_4}$

(iii) $CH_3OH + SOCl_2 \longrightarrow$

24. Construct suitable chemical reactions

(i) CH_3CONH_2 into CH_3NH_2

(ii) $C_6H_5 - NO_2$ into $C_6H_5 - NH_2$

(iii) C_2H_5CN into $C_3H_7NH_2$

25. Calculate the mass of ascorbic acid ($C_6H_8O_6$) to be dissolved in 75g of acetic acid to lower its melting point by 1.5°C (K_6 for acetic acid = 2.93KKg/mol)

26. Two liquid mixtures are prepared.

Mixture A: Benzene + Toluene

Mixture B: Ethanol + water

Mixture A shows no change in volume and no change in heat on mixing.

Mixture B shows change in volume and evolution of heat.

Q1. Which mixture is non ideal. State the type of deviation.

Q2. Explain the reason for deviation.

Q3. State Raoult's law.

27. Arrange the following in order of their increasing boiling points.

i) Pentanol, butanol, butan-2-ol, ethanol, propanol, methanol.

ii) Pentanol, butane, ethoxy ethane.

iii) Why is ethanol water soluble, whereas hexanol is not.

28. A student attempts to prepare aniline using Gabriel phthalimide systhesis method.

i) Why this method is not suitable for preparing aniline.

ii) Why this method is meant for preparation of primary aliphatic amine.

iii) Suggest an alternative method to prepare aniline.

(OR)

i) Define Henry's law.

ii) Discuss the importance of Henry's law.

SECTION – D (2x4=8)

Read the passage carefully and answer the questions that follow.

29. A student places RBCs in three different solutions to study the effect of concentration on cell behaviour. When RBCs are placed in solution A, they swell and bursts because water enters the cells. In solution B, the cells remain normal and there is no movement of water across the cell membrane. However when RBCs are placed in solution C, they shrink because water moves out of the cells into the surrounding solution.

1) Identify the type of solution (hypotonic, hypertonic, Isotonic) for A,B and C. **(1M)**

2) What percentage of NaCl solution administered to patients. **(1M)**

3) Explain any two real time example based on this concept. **(2M)**

30. Living systems are made up of various complex biomolecules like carbo hydrates, proteins nucleic acids, lipids etc., proteins and carbohydrates are essential constituents of our food. These biomolecules interact with each other and constitute the molecular logic of life processes. Some simple molecules like vitamins, and mineral salts also play an important role in the functions of organisms.

1) Write a test to confirm the presence of 5 – OH in the glucose structure.

2) Define Anomers.

3) List out the various bonds stabilising the primary, secondary and tertiary structures of proteins.

4) Define Zwitter Ion.

SECTION – E (3x5=15)

31. A) i) Give an example for symmetrical and unsymmetrical ethers.

ii) Convert Propene into Propanol.

iii) How will you prepare Propan-2-ol from Ethanal.

iv) Boiling point ethanol is higher than methoxy methane. Justify.

v) Reaction of Alcohols with Acid Chloride is carried in the presence of pyridine. Give reason.

(OR)

B) How will you prepare phenols from

i) Chlorobenzene

ii) Benzene sulphonic acid

iii) Benzene diazonium salt

iv) Cumene

32. A) i) Why does boiling point of water increases on adding a non-volatile solute?

ii) How boiling point of water be affected if the solute dissociates?

iii) A solute shows higher molecular mass than expected. Give reason.

iv) A solution deviates strongly from Raoult's law. Why does this deviation occur.

v) Vant hoff factor for Benzoic acid in benzene is less than 1. Justify.

(OR)

B) i) Vapour pressure of water at 293K is 18mm Hg. Calculate the vapour pressure of water when 50g of Glucose is dissolved in 250 g of water. **(3M)**

ii) Define mole fraction. **(2M)**

33. i) How does Lucas test help in distinguishing different types alcohols. Explain with suitable reactions. **(3M)**

ii) Write the mechanisms involved in the dehydration of alcohols. **(2M)**

(OR)

B) i) Why is Iodide a better leaving group than chloride.

ii) Why does S_N1 reaction lead to racemization.

iii) Why do tertiary alkyl halides prefer S_N1 reaction?

iv) Define racemisation.

v) Define Zaitsev rule.